Medicinal chemistry of H1-Antihistamine

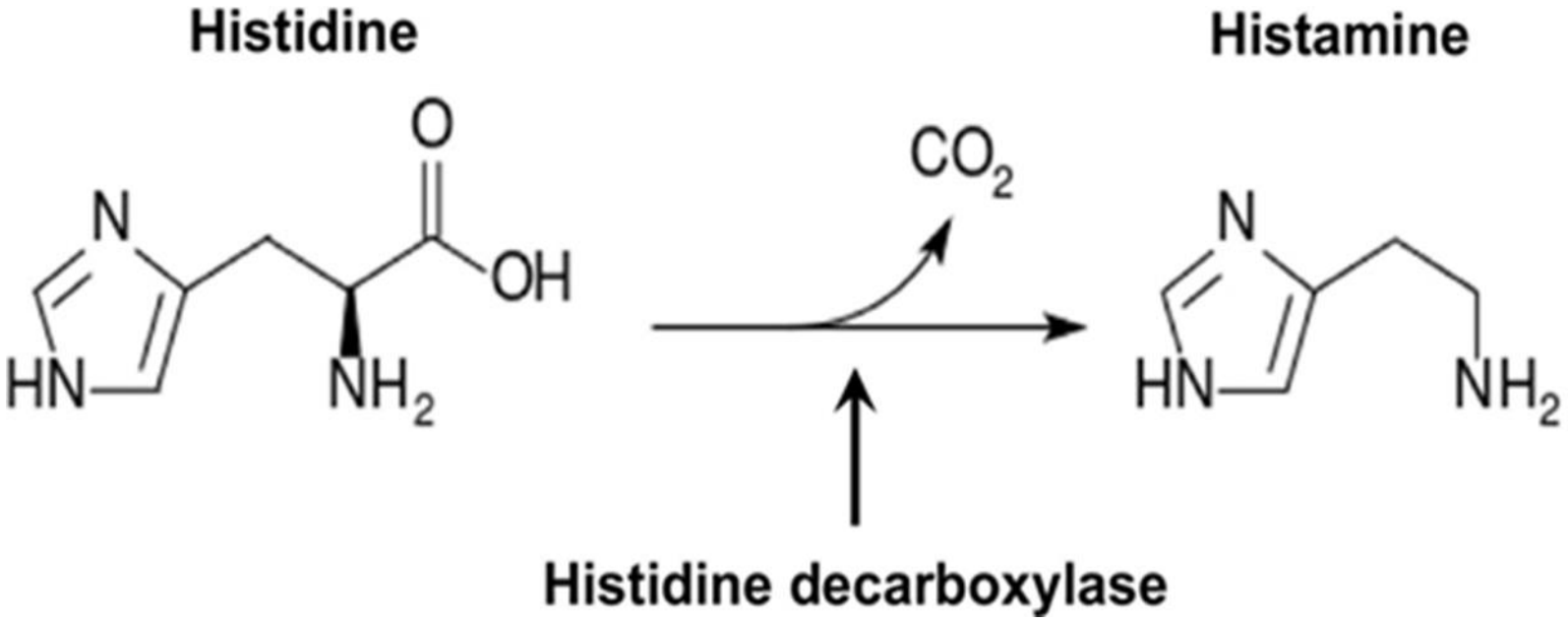
By

Gemmechu H (MSc)

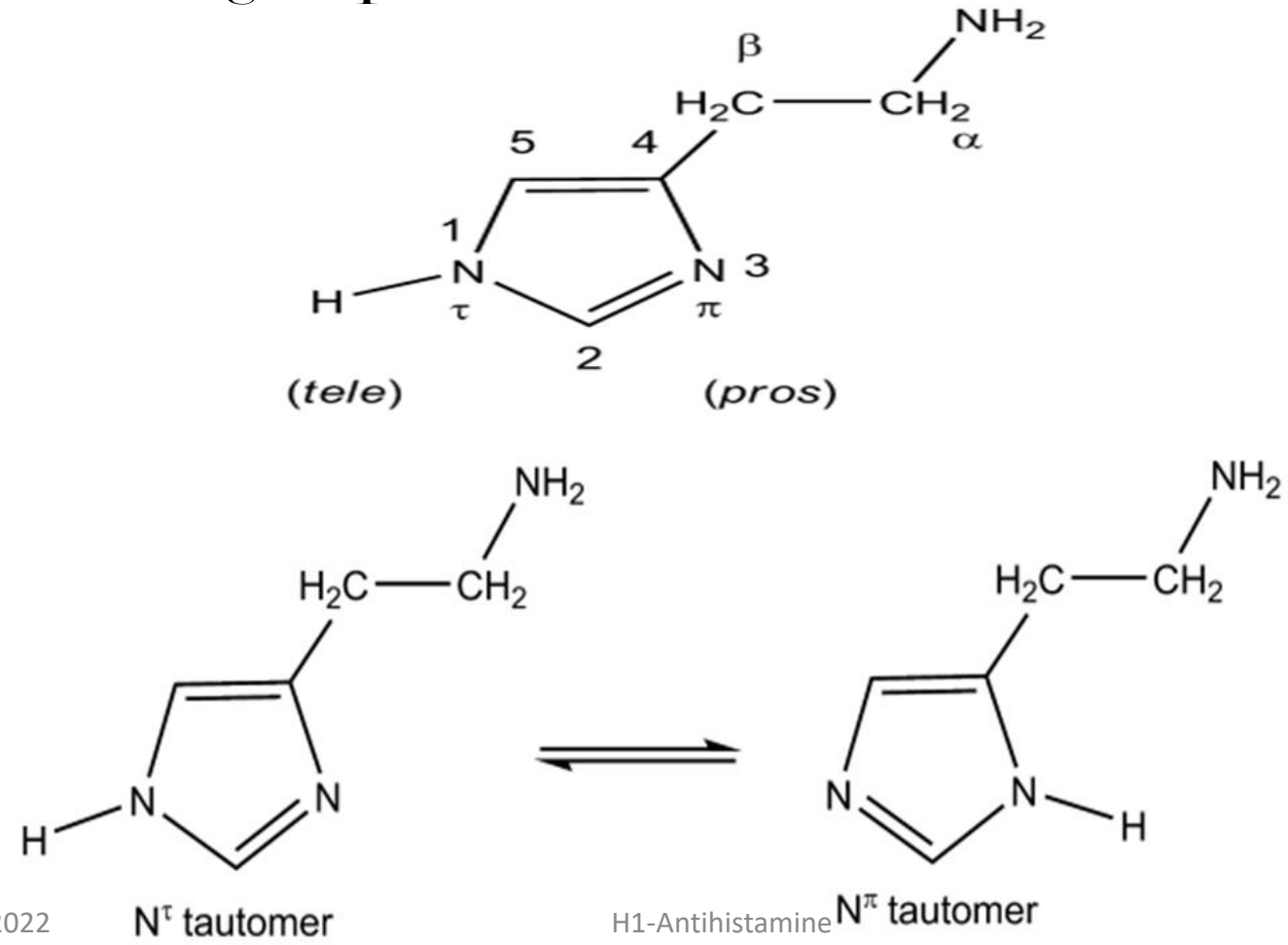
Histamine

- Biogenic amine.
- Produced by mast cells, basophils, ECL cells, histaminergic neurons.
- Play important role and functions:
  - Allergic reactions
  - Inflammatory reactions
  - Gastric acid secretion
  - NT action in brain

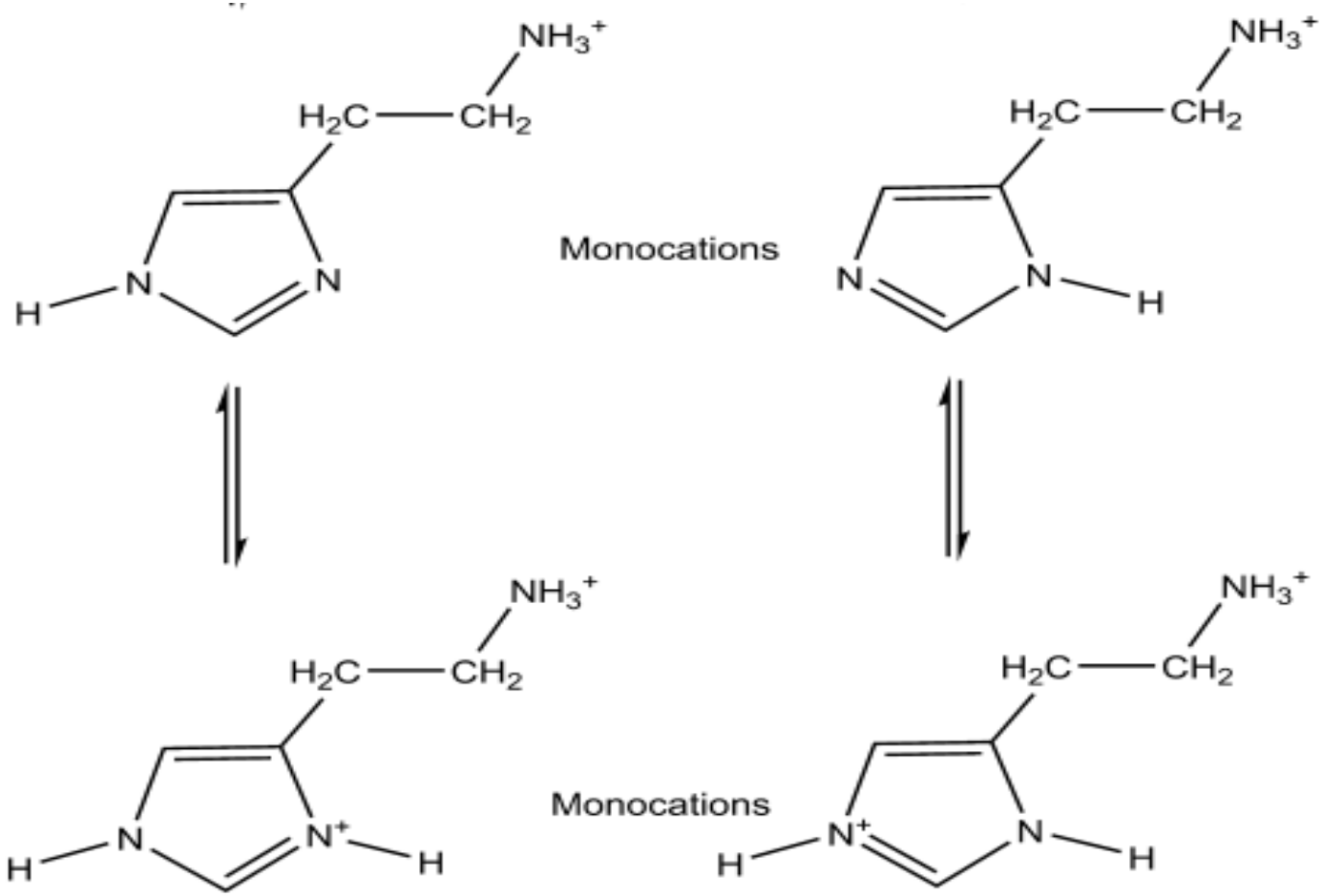
Histamine…


Histidine
Histamine
O
OH
N
HN
NH2
CO2
N
HN
NH2
Histidine decarboxylase

- Made-up of an imidazole ring existing in two tautomeric forms
  - attached to two carbon chain with a terminal α-amino group.



- Histamine exists almost exclusively (96.6%) as the monocationic conjugate species (αNH3+) at physiological pH (7.4).
- At lower pHs, a higher percentage of the dicationic species exists.



Structure–activity relationship

- The two protonated species (monocation and dication).
- The monocation (protonated aliphatic amine) is sufficient for agonist activity.

- Most histamine is biosynthesized and stored as
  - protein complexes in mast cells (complexed with heparin) and
  - basophilic granulocytes (complexed with chondroitin).
- Protein complexed histamine is stored in secretory granules and released by
  - exocytosis in response to a wide variety of immune and non-immune stimuli.

- Once released, the physiological effects of histamine are mediated by specific cell-surface receptors.
- Extensive pharmacological and molecular biology studies have revealed the presence of at least
  - four different histamine receptor subtypes in mammalian systems designated as H_1, H_2, H_3, and H_4

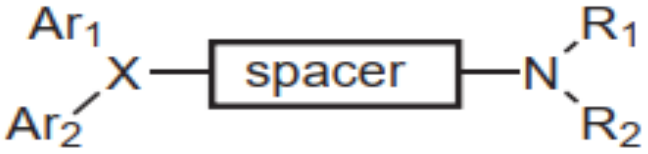
Histamine receptors

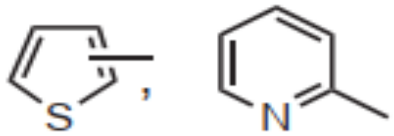
H_1 receptor	• smooth muscle contraction (bronchi, gut, large blood vessels) • Relaxes small arterioles • ↓ peripheral resistance and blood pressure
H_2 receptor	• Gastric acid secretion
H_3 receptor	• Localized on nerve terminals; modulates neurotransmitter release in CNS
H_4 receptor	• Expressed in mast cells, basophils, eosinophils → inflammatory process

Antihistamine

- H1 antihistamines
  - are useful and effective in the treatment of allergic responses.
- Adverse central effects include
  - sedation, drowsiness, decreased cognitive ability, and somnolence.

General structure of H1 antihistamines



Ar_1, Ar_2 = (Y)-Ph-, 

X = N	$R_1 = R_2 = CH_3$
X = CHO	$R_1 - R_2 = (CH_2)_{4-6}$
X = CH	$R_1 = CH_3(H)$, $R_2 = CH_2Ar$

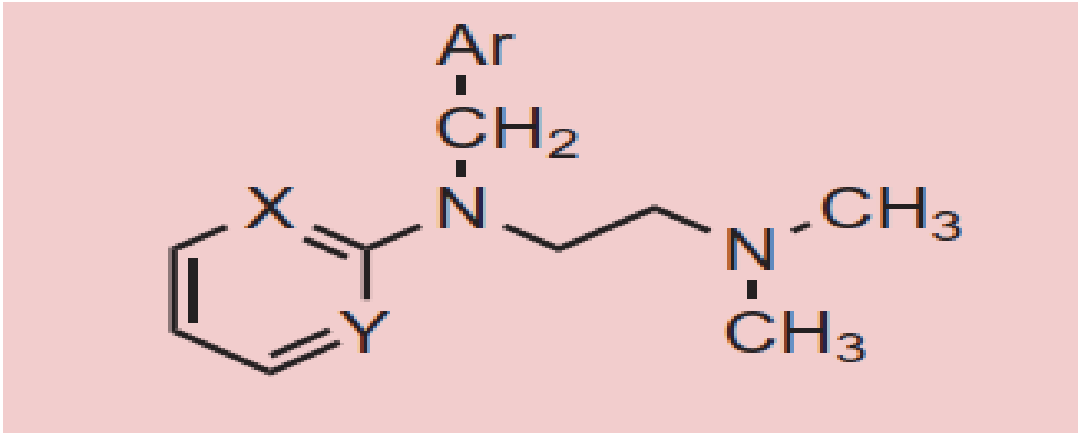
Spacer = $-(CH_2)_n-$
(n = 2 or 3, usually 2)

- The diaryl substitution is essential.
- The co-planarity of two aryl substitutions.
- Most of the antihistamines have ethylene chain, and branching of this chain (inactive).
- The terminal N-atom should be a 3° amine.

First-Generation H1 Antihistamines

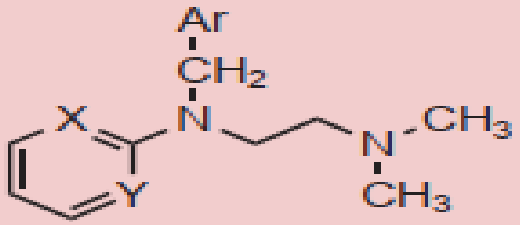
- Ethylenediamines
- Ethanolamine Ethers
- Alkyl Amines
- Piperazines
- Tricyclic H1 Antihistamines

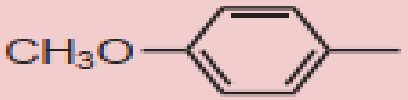
Ethylenediamines

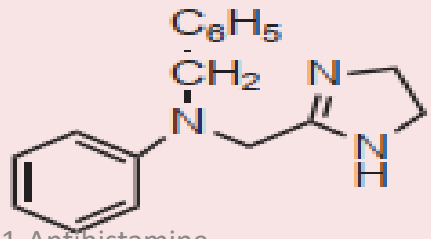


- Aromatic rings are useful (phenyl, 2-pyridyl, halogen- and methoxy-substituted phenyl, or pyrimidyl).
- The small alkyl substituents on basic nitrogen are usually methyl groups
- Exception of antazoline, all have the ethylenediamine spacer.
- In antazoline, the alkyl tertiary amine in phenbenzamine is replaced with an imidazoline group.
- CNS effects, usually sedation, are very common among agents in this class.
- N-demethylation and subsequent deamination.

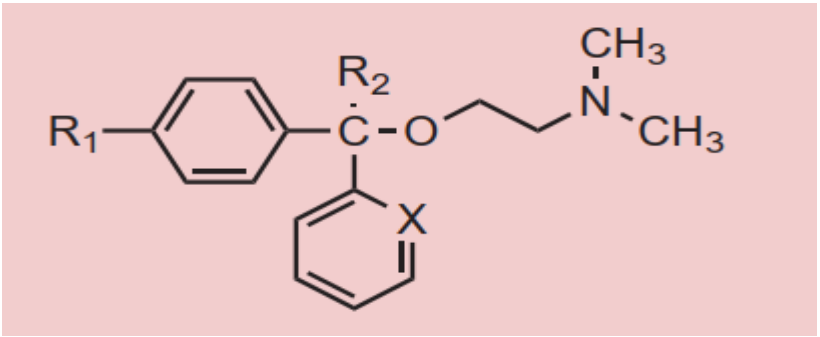
Examples of Ethylenediamine Antihistamines



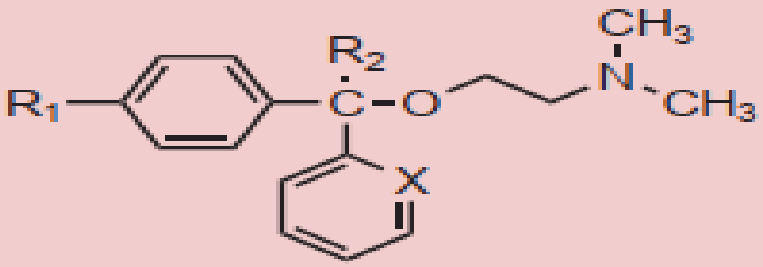
Drugs	X	Y	Ar
Phenbenzamine	CH	CH	
Tripelennamine	N	CH	
Methapyrilene	N	CH	
Thonzylamine	N	N	

Related compound:  Antazoline

Ethanolamine Ethers

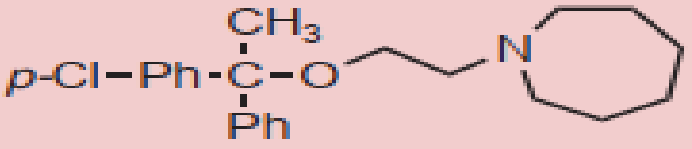


- The compound with the p-Cl-Ph and 2-pyridyl aryl groups is a potent antihistamine (carbinoxamine).
- Substitution of a methyl group at the carbon α to the ether function affords the related compound doxylamine,
  - in which the aryl groups are phenyl and 2-pyridyl.
- Clemastine, a homolog with an additional carbon atom between the oxygen and the basic nitrogen,
  - which is incorporated into a ring, has less sedative properties,
  - as does the drug setastine in which the alkyl amine substituent is incorporated into a seven-membered hexahydroazepine ring.

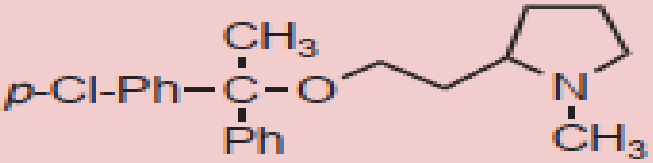


Drugs	Trade Name	R_1	R_2	X
Diphenhydramine Dimenhydrinate	Benadryl Dramamine	H	H	CH
Bromodiphenhydramine		Br	H	CH
Chlorodiphenhydramine		Cl	H	CH
Carbinoxamine		Cl	H	N
Doxylamine	Decapryn	H	CH_3	N
	Unisom			

Related compounds:



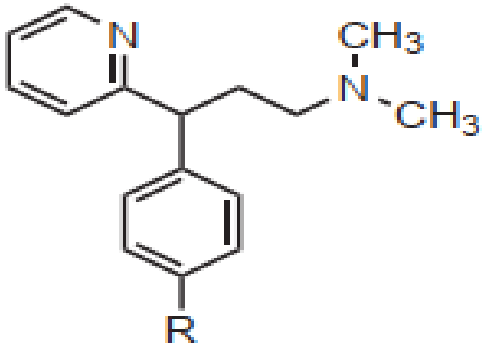
Setastine



Clemastine

Alkyl Amines

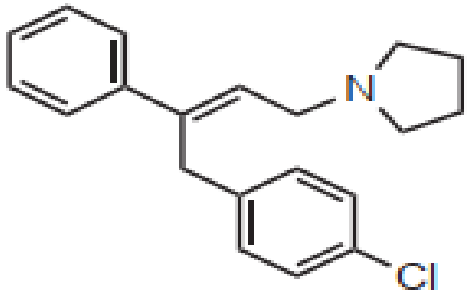
- A carbon atom replaces the heteroatom spacer in the general structure.
- The ring halogen-substituted compounds.
- Characterized by a long duration of action and decreased incidence of central sedative side effects.
- *N*-dealkylation is a major pathway.



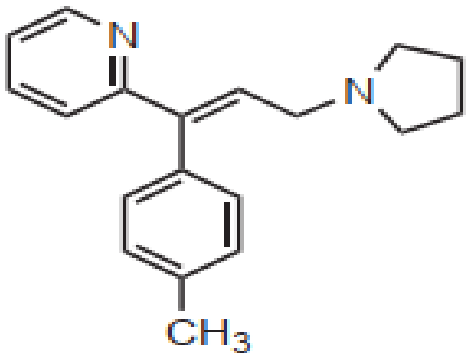
R = H; Pheniramine

R = Cl; Chlorpheniramine
Dexchlorpheniramine

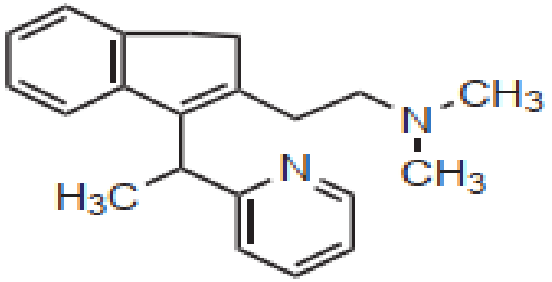
R = Br; Brompheniramine
Dexbrompheniramine



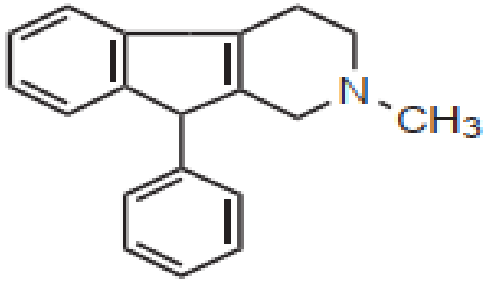
E-Pyrrobutamine



Triprolidine



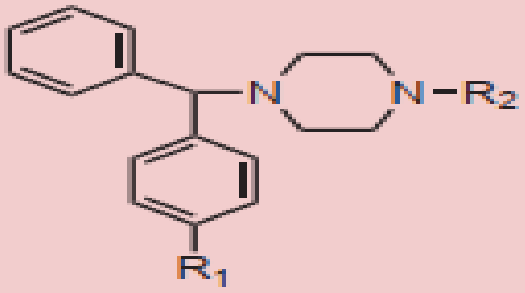
Dimethindene

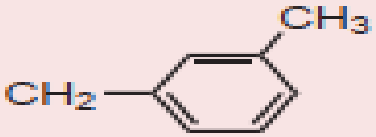
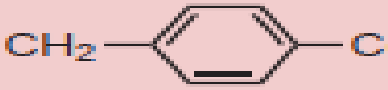
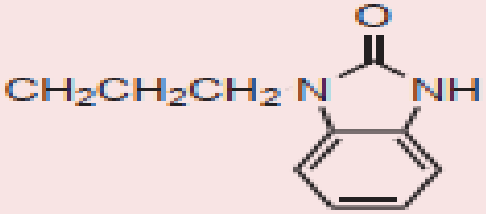


Phenindamine

Piperazines

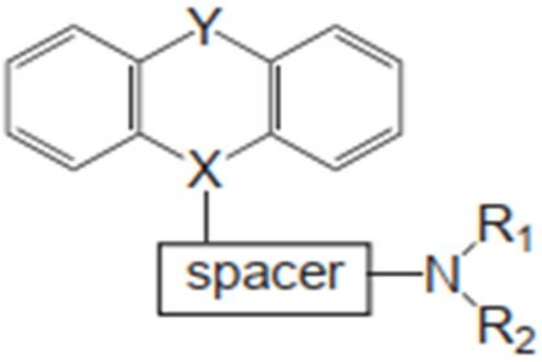
- Members of the piperazine class of agents are structurally related to both
  - the ethylenediamines and
  - The benzhydryl ethers of ethanolamines.
- Their structures include the two-carbon separation between nitrogen atoms,
  - which is incorporated into the piperazine ring.



Drugs	Trade name	R_1	R_2
Cyclizine	Marezine, Bonine	H	CH_3
Chlorcyclizine	Mantadil	Cl	CH_3
Meclizine	Antivert	Cl	
Buclizine	Bucladin-S	Cl	
Oxatomide	Tinset	H	
Hydroxyzine	Vistaril	Cl	$CH_2CH_2OCH_2CH_2OH$
Cetirizine	Zyrtec	Cl	$CH_2CH_2OCH_2COOH$

Tricyclic H1 Antihistamines

- The two aromatic groups noted in several of the classes of antihistamines can be connected to each other
  - Through additional atoms such as the heteroatoms sulfur or oxygen or through a short one- or two-carbon chain.

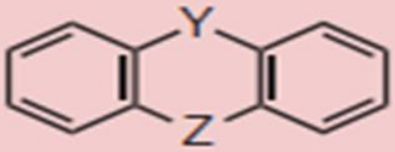


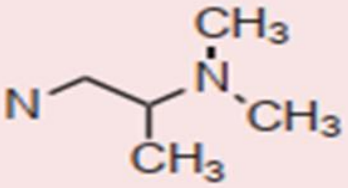
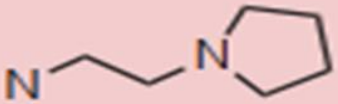
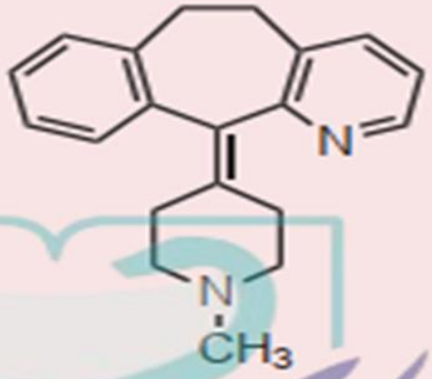
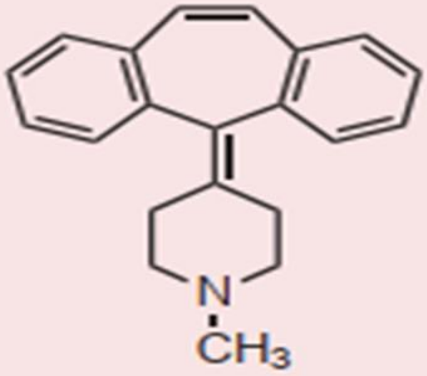
X = C, CH, N, etc.

Y = CH_2, S, O, NH, CH_2O, CH_2CH_2, CH=CH, etc.

spacer = two or three carbons

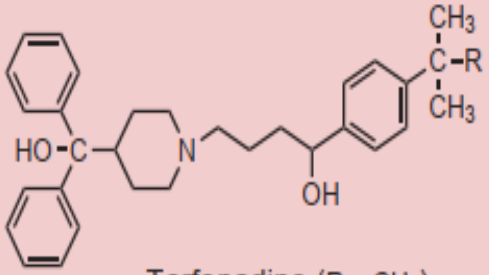
R_1, R_2 = CH_3, or five membered ring



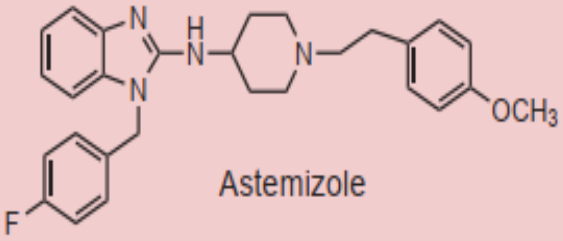
Drugs	Y	Z
Promethazine	S	
Pyrathiazine	S	
 Cyproheptadine		 Azatadine

Second-Generation Non-sedating H1 Antihistamines

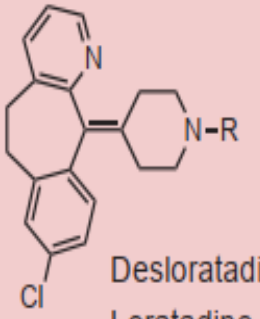
- Have improved H1 selectivity,
  - have little or no sedative qualities, and
  - may have antiallergic effects apart from antihistaminic activity.
- The active agents apparently do not penetrate the blood–brain barrier significantly.



Terfenadine (R = CH_3)
Fexofenadine (R = COOH)

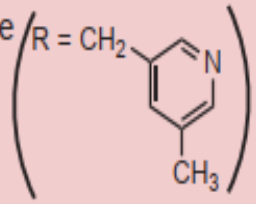


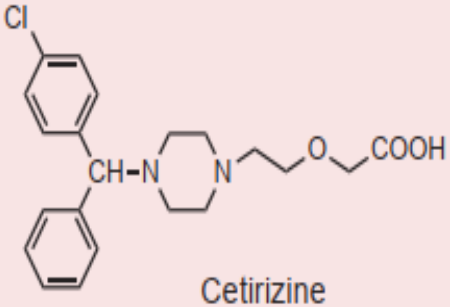
Astemizole



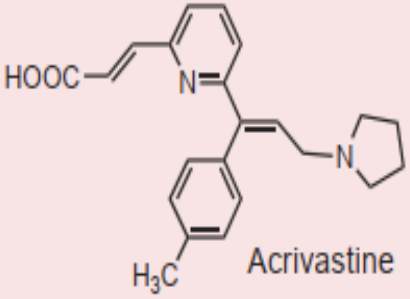
Desloratadine (R = H)
Loratadine (R = $COOCH_2CH_3$)

Rupatadine

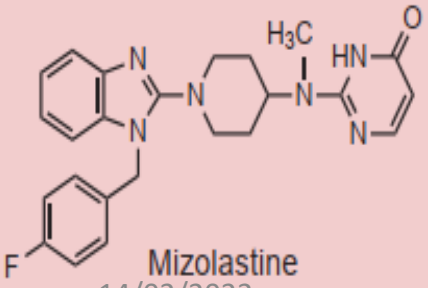




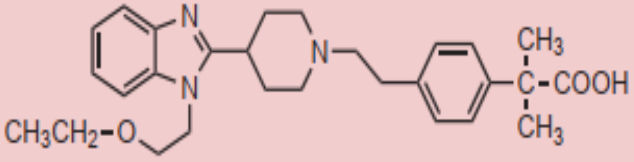
Cetirizine
Levocetirizine



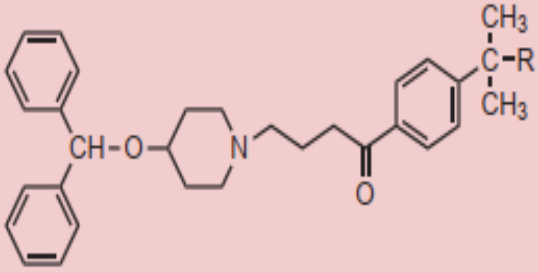
Acrivastine



Mizolastine



Bilastine



Ebastine (R = CH_3)
Carebastine (R = COOH)

•END!